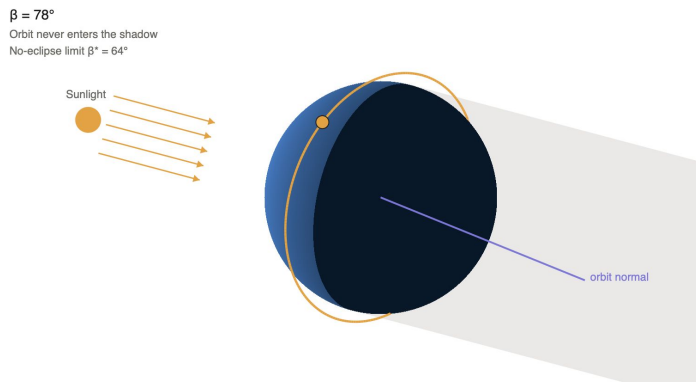
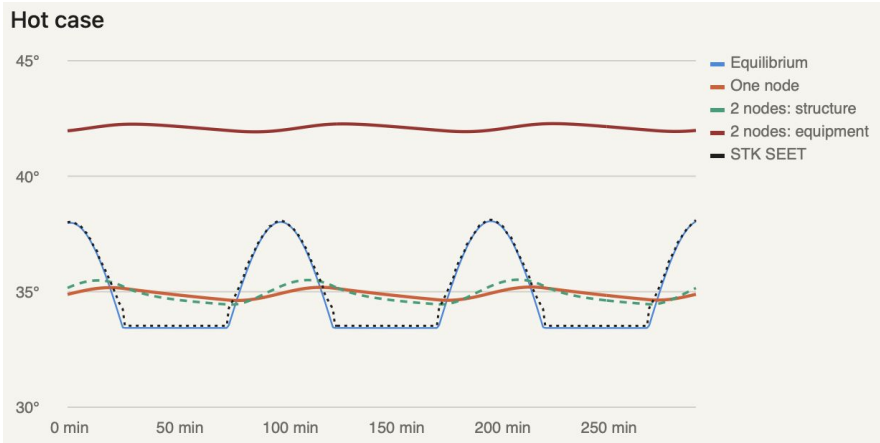
$$T = \left(\frac{Q_{env} + Q_{int}}{\varepsilon \sigma A} \right)^{1/4}, \qquad Q_{int} = 141\,\mathrm{W}$$

$$Q_{env}(t) = \underbrace{\alpha S\, I(t)\, \frac{A}{4}}_{\text{constant}} + \alpha S\, a\, F A\, \max(\cos\beta \cos\theta,\, 0) + \underbrace{\varepsilon\, q_{IR}\, F A}_{\text{constant}}$$

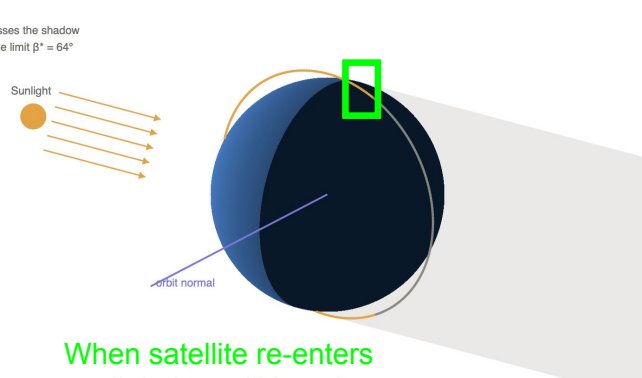


The middle term (albedo) is 0 when satellite enters the night side (of the Earth, not the satellite.) Hence T is constant during that time. This is 50% of the time.



$$T = \left(\frac{Q_{env} + Q_{int}}{\varepsilon \sigma A} \right)^{1/4}, \qquad Q_{int} = 141\,\mathrm{W}$$

$$Q_{env}(t) = \alpha S\, I(t)\, \frac{A}{4} + \alpha S\, a\, F A\, \max(\cos\beta \cos\theta,\, 0) + \underbrace{\varepsilon\, q_{IR}\, F A}_{\text{constant}}$$



The length of the flat bottom depends on the altitude of satellite. If the satellite is higher than the sun gets to lit more of its orbit hence the shorter of the flat bottom part.

